

50 Power BI Interview Questions & Answers

◆ CORE CONCEPTS (Q1–Q10)

1. What is Power BI and its key components? Power BI is Microsoft's business intelligence platform for data visualization and reporting.

Key components:

- **Power BI Desktop** – authoring tool for building reports
- **Power BI Service** – cloud platform for sharing/publishing
- **Power BI Mobile** – iOS/Android consumption app
- **Power Query** – data transformation engine (ETL)
- **DAX** – formula language for calculations
- **Data Model** – in-memory columnar engine

2. Difference between Power BI Desktop, Service, and Mobile

	Desktop	Service	Mobile
Purpose	Build reports	Publish & share	View & interact
Environment	Local app	Cloud (web)	iOS/Android
Data refresh	Manual	Scheduled/automatic	Read-only
Admin features	No	Yes	No
Collaboration	No	Yes	No

3. What is Power Query and how is it used?

Power Query is the ETL (Extract, Transform, Load) engine in Power BI. It uses **M language** behind the scenes. You use it to:

- Connect to 100+ data sources

- Clean and reshape data (remove nulls, split columns, pivot/unpivot)
- Merge and append queries
- Apply reusable transformation steps (recorded in the Applied Steps pane)

It runs before data enters the model, so transformations don't affect report performance.

4. Explain DAX and its basic functions

DAX (Data Analysis Expressions) is a formula language used to create **measures**, **calculated columns**, and **calculated tables**.

Key functions:

- SUM(), AVERAGE(), COUNT(), MIN(), MAX() – aggregations
 - CALCULATE() – modifies filter context
 - FILTER() – returns filtered table
 - IF(), SWITCH() – conditionals
 - RELATED() – fetches value from related table
 - DIVIDE() – safe division (avoids divide-by-zero errors)
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5. What are relationships in Power BI data model?

Relationships connect tables via common key columns. Types:

- **One-to-Many (1:*)** – most common; e.g., one Customer to many Orders
- **Many-to-Many (:)** – requires a bridge table or composite model
- **One-to-One (1:1)** – rare, usually indicates tables can be merged

Relationships have a **cross-filter direction**: Single (default) or Both. Avoid bidirectional unless necessary — it can cause ambiguity and slow performance.

6. Difference between Import, DirectQuery, and Live Connection

Mode	Data stored in Power BI?	Refresh	Best for
Import	Yes (in-memory)	Scheduled	Best performance, most features
DirectQuery	No (queries source live)	Real-time	Large/live datasets
Live Connection	No	Real-time	SSAS, Power BI datasets

Import is fastest.

DirectQuery has DAX/feature limitations.

Live Connection connects to an existing published semantic model.

7. What is a dataflow in Power BI?

A dataflow is a **reusable, cloud-based ETL layer** in Power BI Service. It uses Power Query Online to transform data and stores results in **Azure Data Lake Storage Gen2**.

Benefits:

- Centralizes data prep logic (write once, reuse across reports)
- Reduces repeated transformations in Desktop files
- Supports **computed entities** and **linked entities**
- Enables self-service data engineering for teams

8. How do you create measures vs calculated columns?

	Measure	Calculated Column
Evaluated at	Query time (dynamic)	Refresh time (static)
Context	Filter/row context from visuals	Row context only
Storage	Not stored in model	Stored in model (uses RAM)
Use case	Aggregations, KPIs	Row-level classifications, flags

Rule of thumb: Always prefer measures over calculated columns for aggregations. Use calculated columns only when you need a value per row (e.g., a category label).

9. What are slicers and how do they work?

Slicers are **on-canvas filter controls** that let users interactively filter report visuals. They filter based on cross-filter relationships in the model.

Types: List, Dropdown, Between (range), Relative Date/Time, Tile.

Key features:

- **Sync slicers** – share a slicer state across multiple pages
 - **Slicer panel** – new pane-based slicer experience
 - Can be set to single or multi-select
 - Support "Select All" toggle
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10. Explain bookmarks and drill-through

Bookmarks capture a snapshot of a report page's state (filters, visuals visibility, slicers). Used for:

- Navigation (simulating tabs)
- Show/hide panels
- Storytelling with annotated views

Drill-through allows users to right-click a data point and navigate to a **detail page** filtered to that context. You define drill-through by dragging a field into the "Drill through" filter well on the destination page. Drill-through filters carry over automatically.

◆ SECURITY & ADMINISTRATION (Q11–Q14)

11. What is Row-Level Security (RLS)?

RLS restricts data access at the row level for specific users. Defined in Power BI Desktop using **DAX filter rules** on tables, then roles are assigned in Power BI Service.

Two types:

- **Static RLS** – hardcoded values (e.g., [Region] = "North")
 - **Dynamic RLS** – uses USERPRINCIPALNAME() or USERNAME() to match logged-in user to a data column (scales better)
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12. Difference between Power BI Pro and Premium

	Pro	Premium (Per Capacity)	Premium Per User (PPU)
Cost	~\$10/user/month	Capacity-based	~\$20/user/month

Share with free users	No	Yes	No
Paginated Reports	No	Yes	Yes
AI features	Limited	Full	Full
Deployment pipelines	No	Yes	Yes
Large models	No	Yes	Yes

13. What are gateways and when are they needed?

A **Data Gateway** is a bridge between Power BI Service (cloud) and on-premises data sources (SQL Server, Oracle, files on local network).

Types:

- **Standard (Enterprise) Gateway** – supports multiple users and scheduled refresh
- **Personal Gateway** – single user, not recommended for production

Needed when your data source is behind a firewall or on a local server.
Not needed for cloud sources like Azure SQL, SharePoint Online, or Salesforce.

14. How does Direct Lake mode work?

Direct Lake is a **Microsoft Fabric-exclusive** storage mode. It reads Parquet files directly from **OneLake** (Fabric's data lake) without importing data or running DirectQuery SQL.

It combines the speed of Import mode with the freshness of DirectQuery — no scheduled refresh needed. When data in the lake updates, the

semantic model reflects it immediately. Falls back to DirectQuery if memory limits are hit.

◆ ADVANCED FEATURES (Q15–Q25)

15. What is Copilot in Power BI?

Copilot is Microsoft's **AI assistant** integrated into Power BI (requires Fabric or PPU capacity). Capabilities:

- Generate DAX measures from natural language
- Auto-create report pages from a dataset
- Summarize report data in natural language
- Answer questions about your data using Q&A-style prompts

Available in Power BI Desktop and Service.

16. Explain composite models

Composite models allow a single report to combine **multiple storage modes** — Import, DirectQuery, and Direct Lake — in the same data model. You can also combine data from multiple sources (e.g., a DirectQuery SQL Server table joined with an imported Excel table).

This enables flexibility: keep small lookup tables imported (fast) while large transactional tables stay in DirectQuery (fresh). Introduced to overcome the previous limitation of one mode per model.

17. What are custom visuals and how to import them?

Custom visuals are community or third-party visuals not in the default Power BI library. They are packaged as **.pbviz** files.

Sources:

- **AppSource / Marketplace** — certified visuals (vetted by Microsoft)
- **File import** — any .pbviz downloaded manually

To import: In Desktop → Insert → Get More Visuals → Browse marketplace or import from file. Certified visuals have passed Microsoft's security review; uncertified visuals may not work in some export formats.

18. Difference between visuals and cards

- **Visuals** (charts, matrices, maps) display multi-dimensional data with axes, categories, and series
 - **Cards** display a **single scalar KPI value** — great for headline metrics like Total Revenue or Active Users
 - **Multi-row card** shows multiple KPIs in one visual
 - **New Card visual** (current generation) supports reference labels, trend sparklines, and callout values
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19. What is the role of Paginated Reports?

Paginated Reports (based on SSRS technology) are designed for **pixel-perfect, print-ready documents** — invoices, financial statements, operational reports with many rows.

Unlike standard Power BI reports, they:

- Are designed to span multiple pages
- Support precise layout control
- Can export cleanly to PDF, Word, Excel
- Are authored in **Power BI Report Builder**
- Require Premium/PPU to publish to Service

20. How do you handle large datasets in Power BI?

Strategies:

- Use **aggregations** to pre-summarize large tables
- Enable **incremental refresh** to load only new/changed data
- Use **DirectQuery or Direct Lake** to avoid full import
- Apply **query folding** in Power Query (push transformations to source)
- Use **star schema** to minimize model complexity
- Remove unused columns and rows early in Power Query
- Enable **large dataset storage format** (Premium)

21. What are AI visuals in Power BI?

Built-in AI-powered visuals:

- **Key Influencers** – identifies factors that drive a metric up/down
- **Decomposition Tree** – lets users explore data hierarchically by any dimension
- **Q&A visual** – natural language querying on the canvas
- **Smart Narrative** – auto-generates text summaries of report data
- **Anomaly Detection** – flags unusual data points in line charts automatically

22. Explain incremental refresh

Incremental refresh loads only **new or modified rows** rather than the full dataset on each refresh. Configuration:

1. Define RangeStart and RangeEnd parameters (DateTime type) in Power Query
2. Filter your date column using these parameters
3. Define the refresh policy in Desktop (e.g., keep 3 years, refresh last 3 days)
4. Publish to Premium/PPU workspace

Older historical partitions are preserved; only recent partitions are refreshed, dramatically reducing refresh time and load on source systems.

23. What is the FILTER function in DAX?

FILTER() returns a **table** that is a subset of another table based on a condition.

`FILTER(Sales, Sales[Amount] > 1000)`

It's an **iterator** — it evaluates the condition row by row. Commonly used inside CALCULATE() to apply complex filter conditions that can't be expressed as simple Boolean filters.

Avoid using FILTER on large tables without an indexed column — use Boolean filters in CALCULATE where possible for performance.

24. Difference between ALL and REMOVEFILTERS

Both remove filters from columns/tables, but:

- ALL() can be used as both a **modifier inside CALCULATE** and as a **standalone table function** that returns all rows of a table
- REMOVEFILTERS() can **only** be used as a CALCULATE modifier — its intent is explicitly to remove filters, making code more readable

-- Both equivalent as CALCULATE modifiers:

`CALCULATE([Sales], ALL(Dates[Year]))`

CALCULATE([Sales], REMOVEFILTERS(Dates[Year]))

Prefer REMOVEFILTERS for clarity when your intent is filter removal.

25. What are time intelligence functions?

DAX has built-in functions for date-based calculations — require a **proper Date table** marked as such in the model.

Common ones:

- TOTALYTD(), TOTALQTD(), TOTALMTD() – cumulative totals
- SAMEPERIODLASTYEAR() – same period prior year
- DATEADD() – shift by N periods
- DATESYTD() – returns year-to-date date range
- PREVIOUSMONTH(), PREVIOUSQUARTER(), PREVIOUSYEAR()

Example:

Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Dates[Date]))

◆ DAX DEEP DIVE (Q26–Q28, Q43–Q44)

26. How does CALCULATE work?

CALCULATE is the most important DAX function. It **evaluates an expression in a modified filter context**.

CALCULATE(expression, filter1, filter2, ...)

It does two things:

1. Takes the existing filter context
2. Applies the specified filters (can add, replace, or remove filters)

North Sales = CALCULATE([Total Sales], Region[Region] = "North")

Context transition: when used inside an iterator, CALCULATE converts row context into filter context automatically.

27. What is a star schema and why use it?

A star schema has one central **fact table** (transactions, events) surrounded by **dimension tables** (Date, Customer, Product, Geography).

Why use it in Power BI:

- VertiPaq engine is optimized for it
- Reduces model complexity and ambiguity
- Improves DAX performance (simpler filter propagation)
- Easier to understand for report builders
- Avoids many-to-many relationships

Avoid wide flat tables or snowflake schemas in Power BI — denormalize where possible.

28. Explain Quick Measures

Quick Measures are a **no-code DAX generation** feature. You select a calculation type (e.g., Year-over-Year change, Running total, Filtered value) and map your fields — Power BI writes the DAX measure for you.

Useful for beginners to learn DAX patterns. Access via: Right-click a table → New Quick Measure. The generated DAX is fully editable.

◆ MODELING & DESIGN (Q29–Q38, Q45–Q48)

29. What are workspaces and apps?

- **Workspaces** – collaborative development environments in Power BI Service where teams build and manage reports, datasets, and dataflows. Support roles: Admin, Member, Contributor, Viewer.
 - **Apps** – packaged bundles of dashboards and reports published **from a workspace** for end-user consumption. Apps provide a clean, stable URL and version control — users see the published app while developers iterate in the workspace.
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30. How do you schedule data refresh?

In Power BI Service:

1. Go to the dataset settings
2. Set up a **Data Source Credential** (authenticate to the source)
3. If on-premises, configure a **Gateway** connection
4. Under **Scheduled Refresh**, set frequency (up to 8x/day for Pro, 48x/day for Premium)
5. Optionally configure **refresh failure notifications**

For real-time needs, use DirectQuery or streaming datasets instead.

31. Difference between themes and formatting

- **Themes** – JSON-based templates that define **default** colors, fonts, visual styles across the entire report. Applied globally; ensures brand consistency.
- **Formatting** – **per-visual** overrides applied in the Format pane. Overrides the theme for that specific visual.

Best practice: define a theme for brand colors and fonts, then use per-visual formatting only for exceptions.

32. What is Field Parameters?

Field Parameters allow users to **dynamically swap measures or dimensions** in a visual at runtime using a slicer — without editing the report.

Created via: Modeling → New Parameter → Fields.

Power BI creates a calculated table containing the selected fields. When the user picks a field from the slicer, the visual updates to show that metric or dimension.

Use case: one chart that can show Revenue, Profit, or Units depending on user selection.

33. Explain dynamic titles and labels

Instead of hardcoded titles, you can bind a visual's title to a **measure** that returns text based on slicer selections or filter context.

Dynamic Title = "Sales by " & SELECTEDVALUE('Field Parameter'[Field Parameter], "Metric")

In Format pane → General → Title → set "fx" (conditional formatting) → choose a measure. Same technique applies to data labels, card values, and tooltips.

34. What are decomposition trees?

A decomposition tree is an **AI-assisted visual** that lets users drill into a metric by splitting it across any dimension in any order — interactively chosen at runtime.

Features:

- **AI splits** – Power BI auto-suggests which dimension explains the most variance (High Value / Low Value)

- User can manually choose dimensions at each level
- Locked levels vs. dynamic levels
- Works well for root cause analysis and ad hoc exploration

35. How to optimize Power BI performance?

Model layer:

- Use star schema; avoid wide tables
- Remove unused columns/rows in Power Query
- Use integers for key columns (not strings)
- Avoid bidirectional relationships
- Use aggregation tables for large fact tables

DAX layer:

- Prefer measures over calculated columns
- Use variables (VAR) to avoid repeated evaluation
- Avoid FILTER on large tables; use Boolean filters in CALCULATE
- Avoid row-by-row iterators (SUMX, etc.) on large tables

Report layer:

- Limit visuals per page (aim for <8)
- Use bookmarks to hide/show visuals instead of many pages
- Avoid high-cardinality slicers
- Use Performance Analyzer to identify slow visuals

36. What is the new Fluent 2 visual format?

Fluent 2 is Microsoft's updated design system applied to Power BI visuals, aligning with the broader Microsoft 365 aesthetic. It brings:

- Cleaner, modern visual styling
- Improved accessibility (contrast ratios, focus states)
- Updated default colors and typography
- Consistent look across Power BI and other M365 apps

Visuals can be toggled to the new style; it's progressively rolling out to all built-in visuals.

37. Difference between matrices and tables

	Table	Matrix
Rows	Flat list of rows	Hierarchical row groups
Columns	Fixed columns	Can have dynamic column groupings
Subtotals	No	Yes (row and column totals)
Drill down	No	Yes (expand/collapse hierarchy)
Use case	Detail data	Pivot-style cross-tab analysis

38. What are leader lines in visuals?

Leader lines are the **connecting lines between a data label and its corresponding data point** in pie charts, donut charts, and similar visuals — especially when labels are placed outside the shape.

They prevent ambiguity when labels would otherwise overlap the chart. Configurable in the Format pane for applicable chart types (color, line style, length).

◆ SHARING, EMBEDDING & FABRIC (Q39–Q42)

39. How do you embed Power BI reports?

Two main approaches:

- **Embed for your organization** ("User owns data") – uses Azure AD auth; viewers need Power BI licenses. Used for internal portals via iFrame or Power BI JavaScript SDK.
- **Embed for your customers** ("App owns data") – uses a service principal or master account; no Power BI license required for end users. Requires Premium/Embedded (A SKU) capacity. Used for ISV/SaaS products.

The **Power BI Embedded (Azure)** service provides dedicated capacity for customer-facing embedding.

40. What is Fabric integration with Power BI?

Microsoft Fabric is an end-to-end analytics platform that **unifies Power BI with data engineering, data science, real-time analytics, and data warehousing** — all on top of OneLake.

In Fabric, Power BI semantic models can use **Direct Lake** mode to query OneLake data directly. Fabric workspaces replace Power BI Premium workspaces. The Fabric capacity covers all workloads. Power BI is effectively the **reporting and BI layer** of Microsoft Fabric.

41. Explain calculation groups

Calculation groups are a **modeling feature** (created in Tabular Editor or DAX Studio) that reduces measure proliferation. They group related calculation logic (e.g., time intelligence patterns) into a single table with **Calculation Items**.

Example: Instead of writing 10 measures (Sales MTD, Sales YTD, Sales LY MTD...), you write one base measure + a Calculation Group with items like

"MTD", "YTD", "vs LY". The selected item dynamically modifies the measure's evaluation.

Requires Premium or can be authored in Tabular Editor for any model.

42. What are smart narratives?

Smart Narratives is a built-in AI visual that **auto-generates a natural language text summary** of your report or selected visual. It highlights key trends, outliers, and totals in plain English.

The narrative is **dynamic** — it updates as filters are applied. You can also edit the text and insert **dynamic values** (DAX-backed numbers) directly into the narrative for customized storytelling.

43. Difference between SELECTEDVALUE and VALUES

	SELECTEDVALUE()	VALUES()
Returns	Scalar (single value or alternate result)	Table (one column)
Multiple values	Returns alternate result (default: BLANK)	Returns all values
Use case	Dynamic titles, single-select slicers	Iterating or checking selection

-- Safe for titles:

Title = "Sales for " & SELECTEDVALUE(Region[Region], "All Regions")

44. How do you debug DAX queries? Tools and techniques:

- **DAX Studio** – free tool to write, run, and profile DAX queries; shows server timings and query plans

- **Performance Analyzer** (in Desktop) – shows DAX query time per visual
- **EVALUATE** keyword – run DAX table expressions directly in DAX Studio
- **Variables (VAR)** – break complex measures into testable steps
- **Error trapping with IFERROR()** – catch unexpected BLANKs or errors
- Check **filter context** using FILTERS() or SELECTEDVALUE() debug measures

◆ REMAINING ADVANCED (Q45–Q50)

45. What is the role of Power BI datasets (Semantic Models)?

Datasets (now called **Semantic Models** in Fabric/Power BI) are the **published data models** in Power BI Service — containing tables, relationships, measures, and RLS.

Multiple reports can connect to one shared semantic model (**thin reports**), ensuring a single source of truth. **Endorsed datasets** (Certified/Promoted) signal trustworthiness to other teams. With Fabric, they become first-class items alongside lakehouses and warehouses.

46. Explain what-if parameters

What-if parameters create a **user-controlled numeric slicer** that feeds into DAX measures for scenario analysis.

Created via: Modeling → New Parameter → Numeric Range. Power BI creates a calculated table and an auto-measure (Parameter Value). You reference this in measures:

Adjusted Revenue = [Total Revenue] * (1 + [Discount Rate Value])

Use cases: pricing scenarios, growth rate modeling, budget planning.

47. What are custom tooltips?

Custom tooltips are **full report pages** that appear on hover over a data point — replacing the default tooltip box with a rich, designed page containing any visuals, KPIs, or context.

To create: Build a report page → set it as a Tooltip page (Page settings) → in the visual's Format pane, set Tooltip → Page → select your tooltip page. Supports filtering to the hovered data point automatically.

48. How does AI split column work?

In Power Query, **AI-powered split column** uses machine learning to detect patterns and intelligently split text columns — useful for inconsistently formatted strings (e.g., "John Smith - Manager, New York") where a simple delimiter won't work.

Part of the broader **AI Insights** in Power Query that also includes **Text Analytics** (sentiment, key phrases) and **Azure Machine Learning** model scoring — available with Premium capacity.

49. What is translytical querying?

Translytical (Transactional + Analytical) refers to running **analytical queries directly on operational/transactional data** without ETL delays — blurring the line between OLTP and OLAP.

In the Microsoft Fabric context, **mirroring** brings data from operational sources (Azure SQL, Cosmos DB, Snowflake) into OneLake in near real-time, enabling Power BI to run analytics on fresh transactional data via Direct Lake without a full ETL pipeline.

50. How would you migrate Tableau to Power BI?

Key steps:

1. **Audit** – inventory all Tableau workbooks, data sources, calculated fields, and user permissions
 2. **Data layer** – recreate Tableau data sources as Power BI semantic models; map Tableau calculations to DAX equivalents
 3. **Visuals** – rebuild dashboards in Power BI Desktop (no direct converter; manual rebuild is usually needed for quality)
 4. **Calculated fields** – translate Tableau's LOD expressions and table calculations to DAX (most complex step)
 5. **Security** – re-implement row-level filters as RLS in Power BI
 6. **Governance** – set up workspaces, apps, and endorsement to match Tableau Server/Cloud structure
 7. **Training** – upskill users on Power BI Service, slicers, and report interaction patterns
 8. **Validation** – compare numbers between both platforms before decommissioning Tableau
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